

Mid Question Answer

Terminology →

System →

A system is defined to be collection of entities that act and interact together toward the accomplishment of some logical end.

OR

A system is a collection of components wherein individual components are constrained by connecting interrelationships such that the system as whole fulfills some specific functions in response to varying demands.

Model →

A model is a simplified representation of a real-world system, process or phenomenon that is designed to mimic the behavior of a actual system under different conditions.

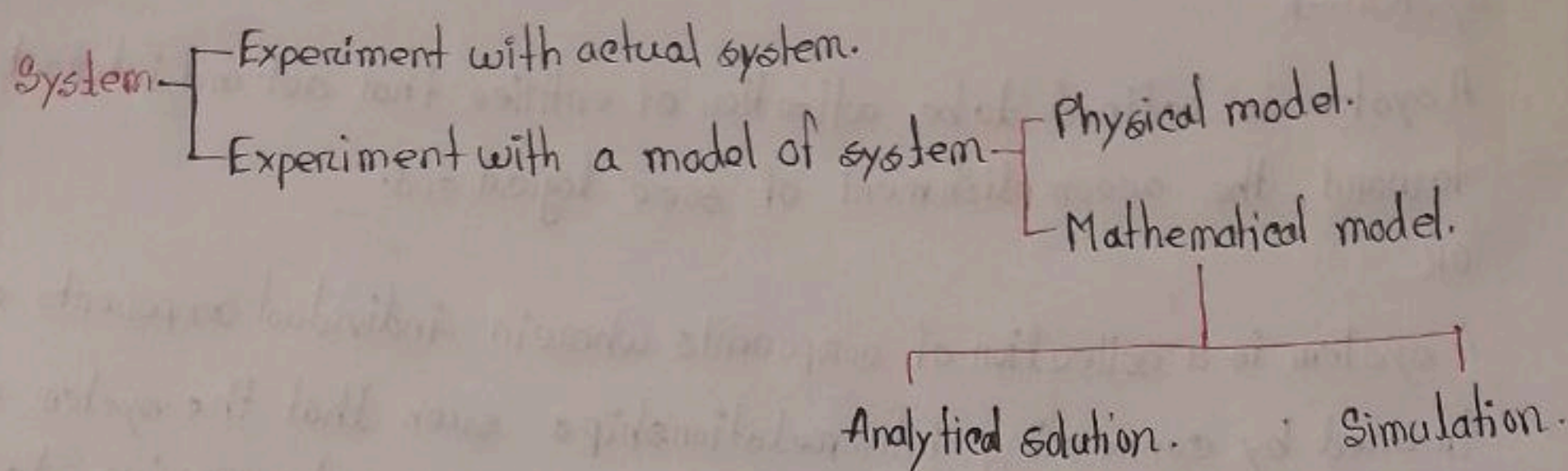
Simulation →

Simulation is the process of imitating the behavior of a real world system over time using models. that allows for ~~experimenting~~ experimentation and analysis without directly interacting with the actual system.

Analytical solution →

An analytical solution is a closed-form mathematical expression that directly solves a given problem using formulas, equations or algebraic manipulation that provides an exact answer rather than an approximation.

Different ways in which system might be studied →



Steps of simulation for a single^{server} queueing system →

Steps →

1. Problem statement
2. Intuitive Explanation
3. Program organization and logic
4. C program
5. Simulation output and discussion
6. Alternative stopping rules
7. Determining events and variables

Describe the Problem statement and intuitive explanation of a single-server queueing system.

Problem statement →

Consider a single-server queueing system for which the ^{interarrival} times $A_1, A_2, \dots$ are independent and identically distributed (IID) random variable.

A customer who arrives and find the server idle enters the service immediately, and the service time $s_1, s_2, \dots$ of the successive customers

are IID random variables that are independent of interarrival time. A customer who arrives and find the server busy joins the end of a single queue. After completing service for a customer, the server chooses a customer from the queue in first in-first out manner (FIFO). The simulation will begin in the "empty-and-idle" state.

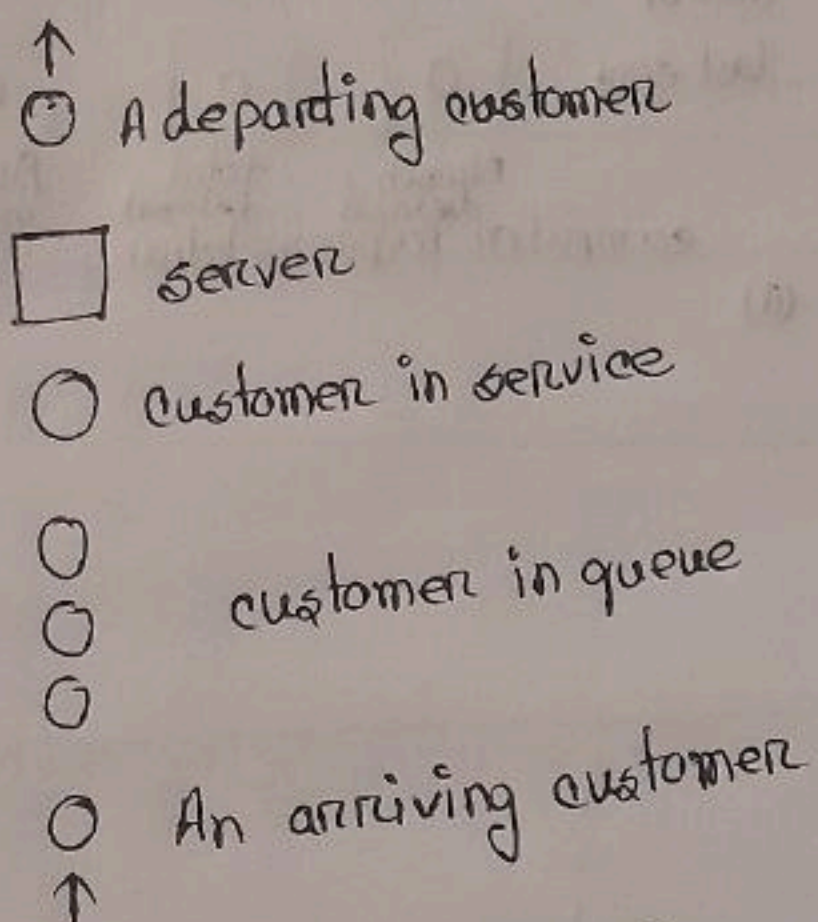


fig → Single-server queueing system.

There are three quantities to measure the performance of the system →

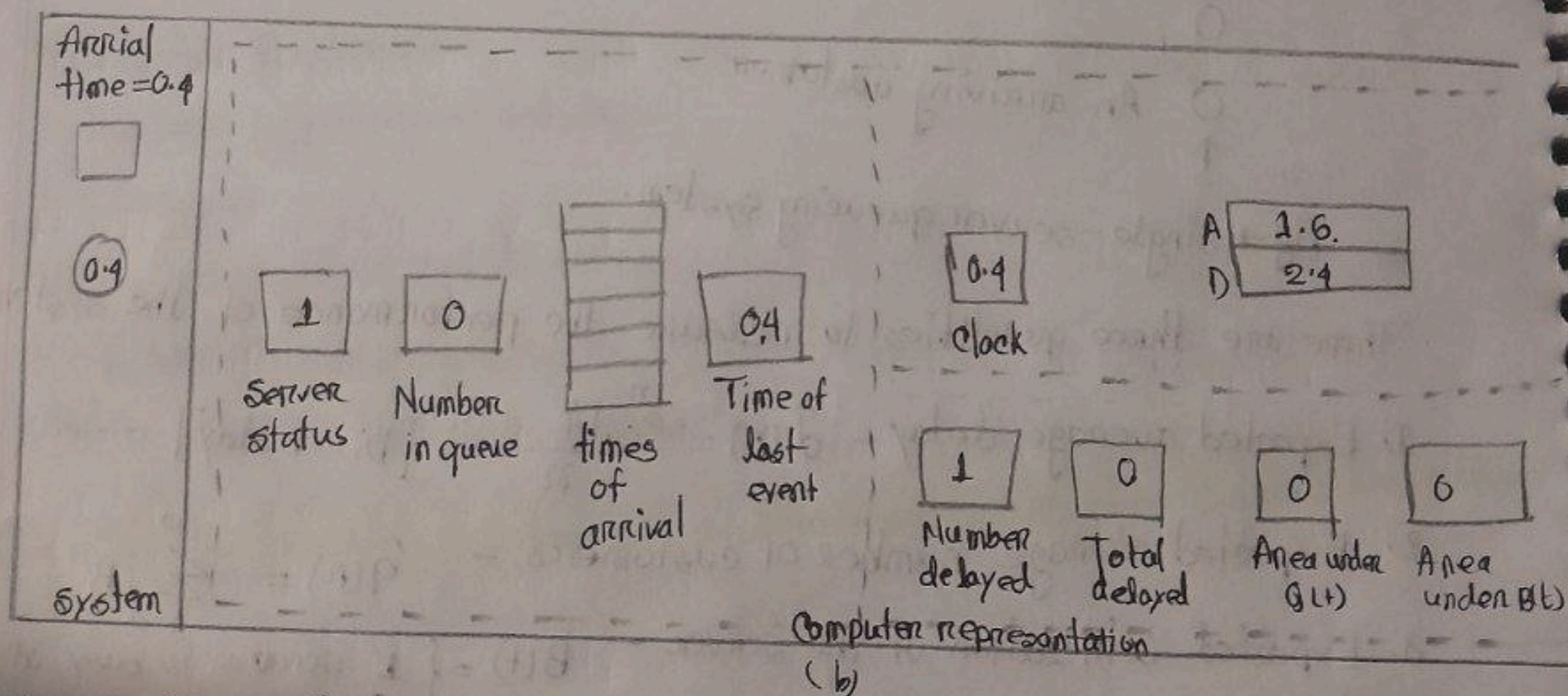
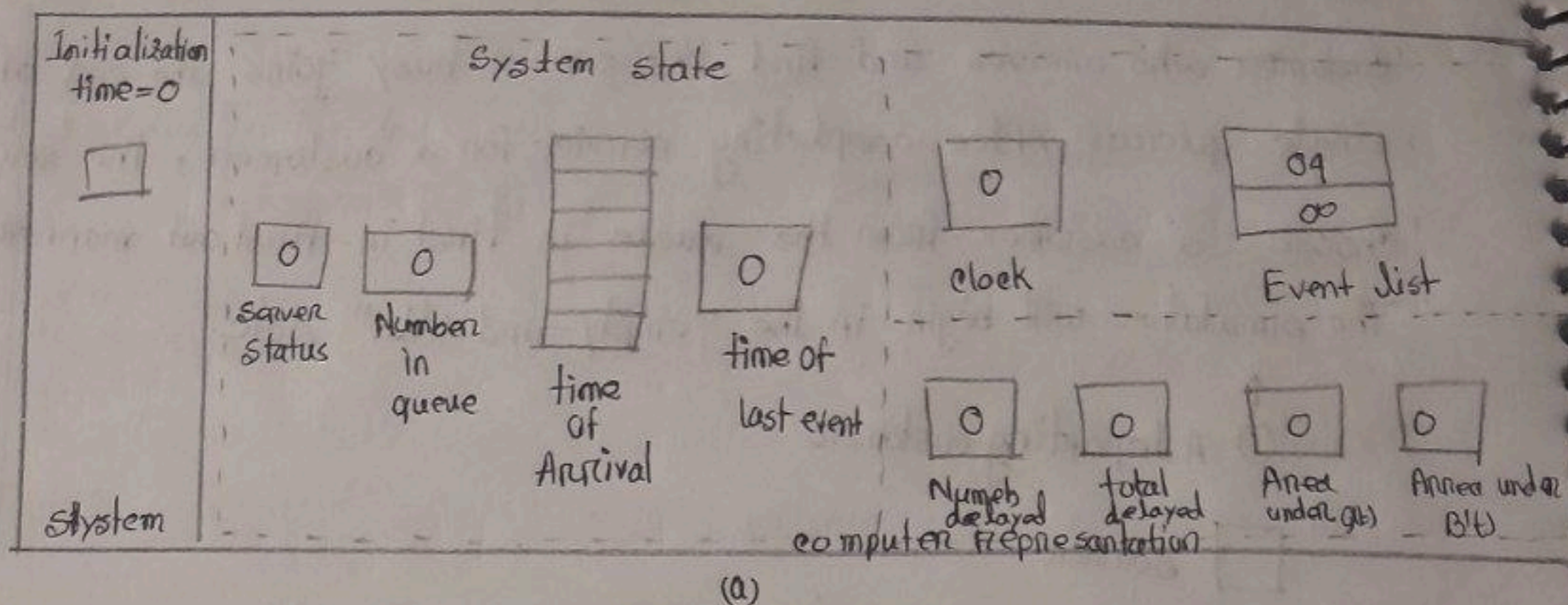
1. Expected average delay - $\hat{d}(n) = \frac{\sum_{i=1}^n D_i}{n}$ [D_i = Delay]
2. Expected average number of customers - $\hat{q}(n) = \sum_{i=0}^{\infty} i P_i$
3. Expected utilization of the server - $B(t) = \begin{cases} 1 & \text{server is busy at time } t \\ 0 & \text{server is idle at time } t \end{cases}$

Intuitive Explanation →

We assume that the interarrival service and service time of customers are -

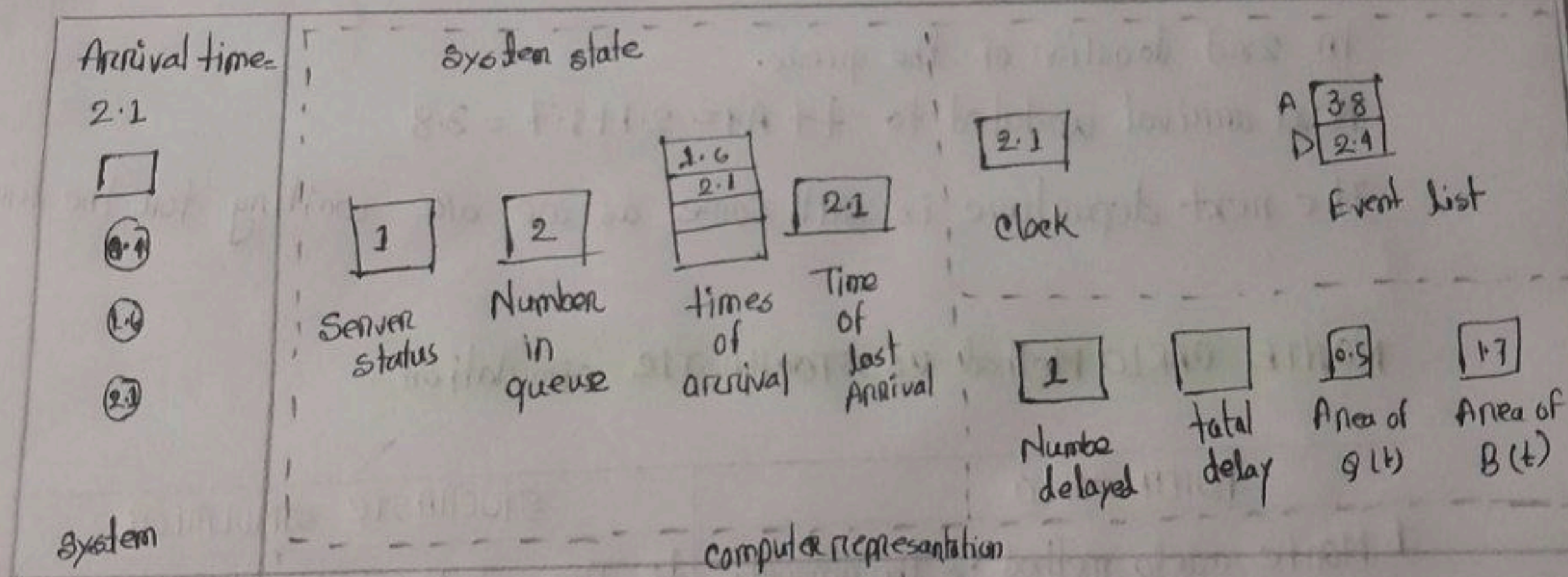
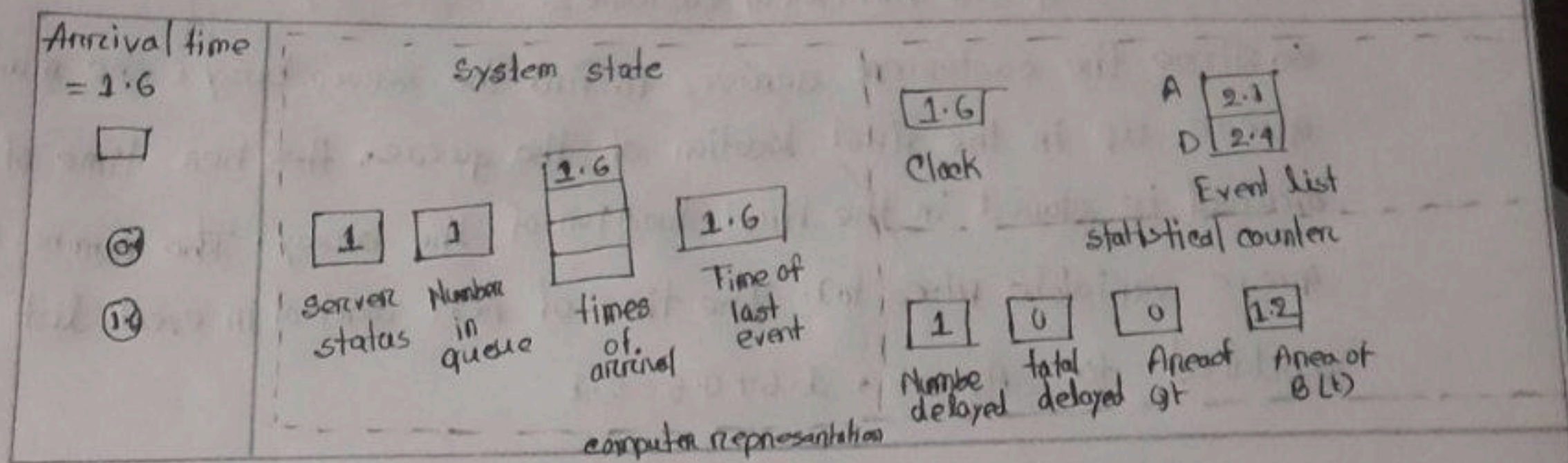
$$A_1 = 0.4, A_2 = 1.2, A_3 = 0.5, A_4 = 1.7, \dots$$

$$B_1 = 2, S_2 = 0.7, S_3 = 0.2, S_4 = 1.1, \dots$$



$t=0 \rightarrow$ [fig-(a)]

Initialization. Initially the system is empty of customer and server is idle. Server status is zero. The number of customers in the queue is 0. There is a one dimensional array to store the time of arrival of the customer.



$t = 0.4 \rightarrow$ [fig (b)] Arrival of customer 1.

At time 0.4 the first customer arrived. Since the server is idle it begins service immediately. The server status set to 1.

Delay in queue $D_1 = 0$. The clock has been advanced to the current time 0.4.

The next arrival will be $A_2 = 1.2$ at the time $1.2 + 0.4 = 1.6$.

The next departure will be $0.4 + 2.0 = 2.4$. The number delayed is increased to one 1.

$t = 1.6 \rightarrow$ [fig-(b)] Arrival of customer 2.

Since the customer arrives to find the server busy, she must queue up in the first location of the queue. Her time of arrival is stored in the first location of the array. The number in queue variable rises to 1. The time of next arrival in event list updated to $A_3 = 0.5, 1.6 + 0.5 = 2.1$

$t_2 = 2.1 \rightarrow$ [fig-(d)] Arrival of customer 3.

The queue grows by one customer, whose time of arrival is stored in 2nd location of the queue.

Next arrival updated to $t + A_4 = 2.1 + 1.7 = 3.8$

The next departure is still same as we are waiting for the service.

MONTÉ CARLO Method vs STOCHASTIC simulation

MONTÉ CARLO	STOCHASTIC SIMULATION
1. Monte carlo method is the computation, which make use of random number to find a solution to problem, which are inherently <u>deterministic</u> in nature.	1. The solution where random number are used for solving problem, which are inherently <u>stochastic</u> in nature are called stochastic simulation.
Example - irregular area of an irregular figure, numerical integration, Evaluation of P, trajectory computation.	Example - The gambling game, bombing problem.

Analytic solution vs simulation

Analytic solution	Simulation
1. Give exact, close-form, mathematical solution	1. Give Approximate, numerical solution using computation
2. Use algebra, calculus, and exact equation	2. Use iterative computational method like Monte Carlo, finite element.
3. Exact and deterministic	3. Approximate and depends on numerical precision
4. Minimal once derived	4. Can be computationally expensive.
5. Limited to solvable equation	5. Can handle complex, non-linear or chaotic system.

Explain next time advance approach.

Two principle approaches have been suggested for advancing the simulation clock

1. Next event time advance
2. fixed-increment time advance.

Next event time advance approach-

With the next event time advance approach, the simulation clock is initialized to zero and the time of ~~see~~ event of future are determined. The simulation clock is then advance to the time of first event of these future events at which point the state of the system is updated to account for the

To illustrate the case of a random process, let us consider a game in which an unbiased coin is repeatedly flipped. For each flip you have to pay Rs. 1 and when the difference between heads and tails tossed becomes 3, you get Rs. 8. Thus if the required difference is obtained in less than 8 flips, you win some money and if it is in more than 8 flips you lose. How to decide, whether to play this game or not?

The easiest way to simulate this game is to use a random number table.

fact that an event has occurred, and our knowledge of time of occurrence of future events also updated. Then the simulation clock advance to the time of new event. This process continued until eventually some prespecified stopping condition is satisfied.

Describe steps in simulation study.

Steps in simulation study-

- Problem formulation
- Model construction
- Data collection
- Model programming
- Validation
- Design of experiment
- Simulation run and analysis
- Documentation
- Implementation

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Continuous system simulation-

Simulation of a continuous system refers to the process of modeling and analyzing a system where variables change continuously over time rather than discrete steps.

Disadvantage of analog simulation-

Analog simulation-

Analog simulation, which uses physical or electrical analog models to represent real world system.

Disadvantage-

1. Limited Accuracy leading to lower precision compare to digital simulation.
2. Lack of flexibility which is modify to simulate different ~~sim~~ scenarios is difficult and time consuming.
3. Scaling issues to scale up for large, complex system.
4. Higher sensitive to Environmental factors like temperature, humidity etc.

Random variable-

A random variable is a numerical value associated with the outcomes of a random experiment.